ORBITALQUERY

Earth Observation Analysis Report

HYDERABAD

28 Feb 2018 > 21 Feb 2026

ANALYSIS SNAPSHOT

DETECTED REGIONS	CHANGED AREA	MEAN |ΔNDVI|	MEAN NDVI
250	40.5 km²	0.285	0.241 > 0.155

ANALYSIS CONTEXT

Location	Hyderabad
Analysis	urban change
Dataset	sentinel-2-l2a
Date range	2018-01-01 to 2026-12-31
AOI area	471 km²

KEY FINDING

Vegetation signal decreased across the analysed area. Mean NDVI changed from 0.241 to 0.155, with 40.5 sq km exceeding the configured |dNDVI| >= 0.20 threshold.

OBSERVATION DATA

1. Search

Query	Hyderabad
Location	Hyderabad
Analysis type	urban change
Date range	2018-01-01 to 2026-12-31
Dataset	sentinel-2-l2a
AOI area	471 km²

2. Selected Scenes

	BEFORE	AFTER
Acquisition	28 Feb 2018	21 Feb 2026
Platform	S2A	S2B
Cloud cover	0.0%	0.0%
Tile	44QKE	44QKE
Dataset	sentinel-2-l2a	sentinel-2-l2a

Scene IDs

Before: S2A_MSIL2A_20180228T050741_R019_T44QKE_20201014T014802

After: S2B_MSIL2A_20260221T050729_R019_T44QKE_20260221T104107

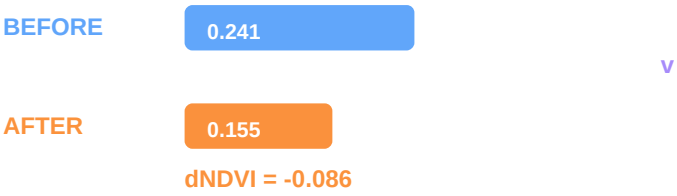
3. Satellite Imagery

Satellite imagery could not be embedded (cross-origin restriction). View the interactive console for full imagery.

CHANGE ANALYSIS

Metric	Value
Detected regions	250
Changed area	40.5 km²
Mean magnitude	0.285
Largest region	6.12 km²
AOI area	471 km²
Threshold	|dNDVI| >= 0.20
NDVI before (mean)	0.241
NDVI after (mean)	0.155

NDVI Comparison

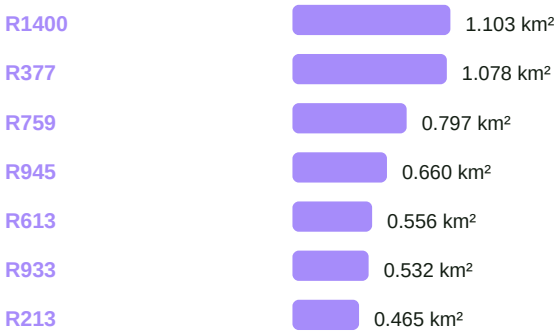


Vegetation signal decreased across the analysed area. Mean NDVI changed from 0.241 to 0.155, with 40.5 sq km exceeding the configured |dNDVI| >= 0.20 threshold.

Top 10 Detected Regions

Rank	Region	Area (km²)	|dNDVI|	NDVI Before	NDVI After
1	R272	6.125	0.384	0.378	-0.004
2	R1016	3.755	0.401	0.376	-0.021
3	R595	1.104	0.580	0.070	0.590
4	R1400	1.103	0.400	0.594	0.195
5	R377	1.078	0.276	0.509	0.234
6	R759	0.797	0.340	0.445	0.107
7	R945	0.660	0.340	0.403	0.063
8	R613	0.556	0.355	0.480	0.137
9	R933	0.532	0.332	0.359	0.027
10	R213	0.465	0.307	0.380	0.074

Top Change Regions by Area


R1400
1.103 km²
R377
1.078 km²
R759
0.797 km²
R945
0.660 km²
R613
0.556 km²
R933
0.532 km²
R213
0.465 km²

METHOD AND INTERPRETATION

1. Method

- Dataset: Sentinel-2 Level-2A imagery retrieved through Microsoft Planetary Computer.
- Vegetation index: NDVI = (B08 - B04) / (B08 + B04)
- Change metric: |dNDVI| = |NDVI_after - NDVI_before|
- Detection threshold: |dNDVI| >= 0.20
- Processing: AOI clipping, valid-pixel masking, morphological cleanup, connected-component filtering, change-region vectorization.

2. Data Provenance

Dataset	sentinel-2-l2a
Provider	Microsoft Planetary Computer (Copernicus Sentinel-2)
BEFORE acquisition	2018-02-28T05:07:41.026000Z
AFTER acquisition	2026-02-21T05:07:29.024000Z
Tile	44QKE
Cloud cover	0.0%
Threshold	|dNDVI| >= 0.20

3. Interpretation

Vegetation signal decreased across the analysed area. Mean NDVI changed from 0.241 to 0.155, with 40.5 km² exceeding the configured |dNDVI| "e 0.20 threshold.

SCIENTIFIC NOTE

> *EO-derived change regions are evidence for review and are not validated ground truth. Results depend on the selected imagery, preprocessing, spatial resolution and configured detection threshold.*